

Table 1: GCI–Richardson extrapolation vs. Roark: deflection and rotation.

Job	Load type	QoI	ϕ_1	GCI ₁₂ (%)	GCI ₂₃ (%)	GCI ₂₃ /GCI ₁₂	p	r^p	$\Delta(\phi_1, \text{Roark})$ (%)	ϕ_{rich}
0	End	u_y (mm)	−3.041	—	—	—	—	—	0	−3.041
0	End	θ_z (deg)	−0.1307	—	—	—	—	—	0	−0.1307
1	Mid-span	u_y (mm)	−0.9504	—	—	—	—	—	0	−0.9504
1	Mid-span	θ_z (deg)	−0.0327	—	—	—	—	—	0	−0.0327
2	Quarter-span	u_y (mm)	−0.2614	—	—	—	—	—	0	−0.2614
2	Quarter-span	θ_z (deg)	−0.0082	—	—	—	—	—	0	−0.0082
3	UDL	u_y (mm)	−2.281	0.003	0.01	4.00	2.00	4.00	0	−2.281
3	UDL	θ_z (deg)	−0.0871	0.004	0.015	4.00	2.00	4.00	0	−0.0871
4	Triangular	u_y (mm)	−1.673	0.002	0.006	4.00	2.00	4.00	0	−1.673
4	Triangular	θ_z (deg)	−0.0653	0.003	0.012	4.00	2.00	4.00	0	−0.0653
5	Parabolic	u_y (mm)	−1.318	0.001	0.006	4.00	2.00	4.00	0	−1.318
5	Parabolic	θ_z (deg)	−0.0523	0.003	0.012	4.00	2.00	4.00	0	−0.0523

Note. Grids: $n_1 = 128$, $n_2 = 64$, $n_3 = 32$ elements; refinement ratio $r = 2$; safety factor $F_s = 1.25$

(three-grid): we use the standard three-grid GCI formula, and the factor 1.25 is the one recommended for that case. ϕ_1 = fine-grid solution; p = apparent order ($p_{\text{th}} = 2$). $\Delta(\phi_1, \text{Roark}) = 100(\phi_1 - \text{Roark})/\text{Roark}$; ϕ_{rich} = Richardson extrapolate. $\text{GCI}_{23}/\text{GCI}_{12} \approx r^p$ in asymptotic range.